

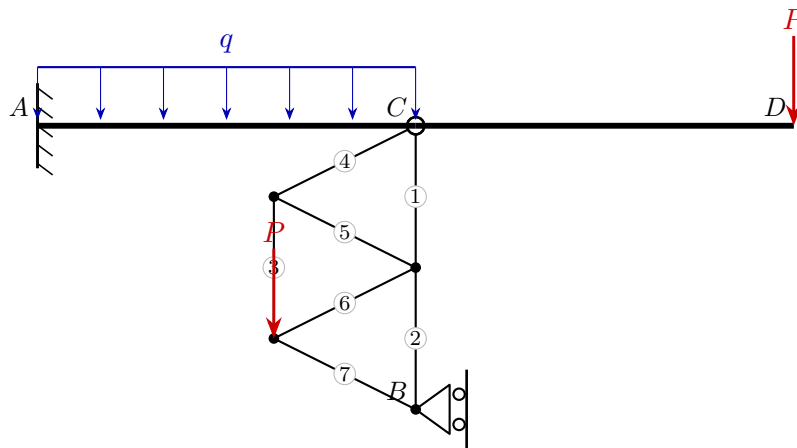
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VU Mechanik – LAWI 100515 (PI)

Exam **Group A**

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Model solution



1. Static determinacy

Counting criterion $n = 3k - (a + z)$ (bodies k , reactions a , interaction forces z):

$$n = 3k - (a + z) = 3 \cdot 2 - (4 + 2) = 0 \Rightarrow \text{statically determinate}$$

2. Support reactions and hinge forces

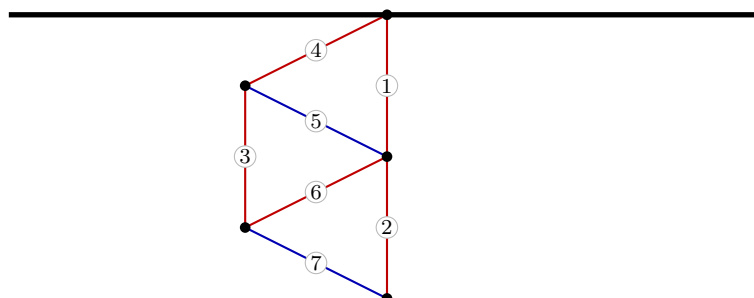
$A: R_x = 5, R_y = 36, M = 152$ $B: R_x = -5, R_y = 0$

$C: H_x = -5, H_y = -10$

3. Truss member forces

Member	Force S [kN]	Type
1	7.5	tension
2	2.5	tension
3	5	tension
4	5.59	tension
5	-5.59	compression
6	5.59	tension
7	-5.59	compression

red: tension (+) blue: compression (-)



4. Section-force diagrams of the beams